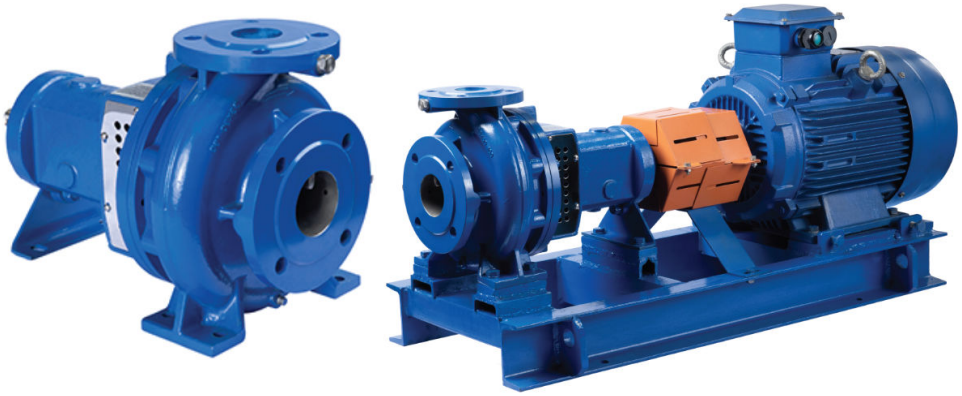
xylem
Let's Solve Water

Installation, Operation, and Maintenance Manual

90036301_1.0



Series 1710

GOULDS WATER TECHNOLOGY
a xylem brand

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1 Introduction and Safety

1.1 Introduction

Purpose of this manual

The purpose of this manual is to provide necessary information for:

- Installation
- Operation
- Maintenance



CAUTION:

Read this manual carefully before installing and using the product. Improper use of the product can cause personal injury and damage to property, and may void the warranty.

NOTICE:

Save this manual for future reference, and keep it readily available at the location of the unit.

1.1.1 Inexperienced users



WARNING:

This product is intended to be operated by qualified personnel only.

Be aware of the following precautions:

- This product is not to be used by anyone with physical or mental disabilities, or anyone without the relevant experience and knowledge, unless they have received instructions on using the equipment and on the associated risks or are supervised by a responsible person.
- Children must be supervised to ensure that they do not play on or around the product.

Requesting other information

The instructions and warnings that are provided in this manual concern the standard version, as described in the sales document. Special versions can be supplied with supplementary instruction leaflets. See the sales contract for any modifications or special version characteristics. For instructions, situations, or events that are not considered in this manual or in the sales documents, please contact the nearest Xylem representative.

Always specify the exact product type and identification code when requesting technical information or spare parts from the sales and service department.

1.2 Safety



WARNING:

- The operator must be aware of safety precautions to prevent physical injury.
- Operating, installing, or maintaining the unit in any way that is not covered in this manual could cause death, serious personal injury, or damage to the equipment. This includes any modification to the equipment or use of parts not provided by Xylem. If there is a question regarding the intended use of the equipment, please contact a Xylem representative before proceeding.
- Do not change the service application without the approval of an authorized Xylem representative.



CAUTION:

You must observe the instructions contained in this manual. Failure to do so could result in physical injury, damage, or delays.

1.2.1 Safety terminology and symbols

About safety messages

It is extremely important that you read, understand, and follow the safety messages and regulations carefully before handling the product. They are published to help prevent these hazards:

- Personal accidents and health problems
- Damage to the product and its surroundings
- Product malfunction

Hazard levels

Hazard level	Indication
 **DANGER:**	A hazardous situation which, if not avoided, will result in death or serious injury
 **WARNING:**	A hazardous situation which, if not avoided, could result in death or serious injury
 **CAUTION:**	A hazardous situation which, if not avoided, could result in minor or moderate injury
NOTICE:	Notices are used when there is a risk of equipment damage or decreased performance, but not personal injury.

Special symbols

Some hazard categories have specific symbols, as shown in the following table.

Electrical hazard	Magnetic fields hazard
 **Electrical Hazard:**	 **CAUTION:**

Hot surface hazard

Hot surface hazards are indicated by a specific symbol that replaces the typical hazard level symbols:



CAUTION:

1.2.2 User safety

General safety rules

These safety rules apply:

- Always keep the work area clean.
- Pay attention to the risks presented by gas and vapors in the work area.
- Avoid all electrical dangers. Pay attention to the risks of electric shock or arc flash hazards.
- Always bear in mind the risk of drowning, electrical accidents, and burn injuries.

Safety equipment

Use safety equipment according to the company regulations. Use this safety equipment within the work area:

- Hard hat
- Safety goggles, preferably with side shields
- Protective shoes
- Protective gloves
- Gas mask
- Hearing protection
- First-aid kit
- Safety devices

Note: Never operate a unit unless safety devices are installed.

Electrical connections

Electrical connections must be made by certified electricians in compliance with all international, national, state, and local regulations. For more information about requirements, see sections dealing specifically with electrical connections.

Precautions before work

Observe these safety precautions before you work with the product or are in connection with the product:

- Provide a suitable barrier around the work area, for example, a guard rail.
- Make sure that all safety guards are in place and secure.
- Make sure that you have a clear path of retreat.
- Make sure that the product cannot roll or fall over and injure people or damage property.
- Make sure that the lifting equipment is in good condition.
- Use a lifting harness, a safety line, and a breathing device as required.
- Allow all system and pump components to cool before you handle them.
- Make sure that the product has been thoroughly cleaned.
- Disconnect and lock out power before you service the pump.
- Check the explosion risk before you weld or use electric hand tools.

Precautions during work

Observe these safety precautions when you work with the product or are in connection with the product:

- Never work alone.
- Always wear protective clothing and hand protection.
- Stay clear of suspended loads.
- Always lift the product by its lifting device.
- Beware of the risk of a sudden start if the product is used with an automatic level control.
- Beware of the starting jerk, which can be powerful.
- Rinse the components in water after you disassemble the pump.
- Do not exceed the maximum working pressure of the pump.

- Do not open any vent or drain valve or remove any plugs while the system is pressurized. Make sure that the pump is isolated from the system and that pressure is relieved before you disassemble the pump, remove plugs, or disconnect piping.
- Never operate a pump without a properly installed coupling guard.

1.2.3 Protecting the environment

Emissions and waste disposal

Observe the local regulations and codes regarding:

- Reporting of emissions to the appropriate authorities
- Sorting, recycling and disposal of solid or liquid waste
- Clean-up of spills

Exceptional sites



CAUTION: Radiation Hazard

Do NOT send the product to Xylem if it has been exposed to nuclear radiation, unless Xylem has been informed and appropriate actions have been agreed upon.

Recycling guidelines

Always follow local laws and regulations regarding recycling.

1.3 Spare parts



WARNING:

Only use original spare parts to replace any worn or faulty components. The use of unsuitable spare parts may cause malfunctions, damage, and injuries as well as void the guarantee.



CAUTION:

Always specify the exact product type and part number when requesting technical information or spare parts from the Sales and Service Department.

1.4 Product warranty

Coverage

Xylem undertakes to remedy defects in products from Xylem under these conditions:

- The faults are due to defects in design, materials, or workmanship.
- The faults are reported to a local sales and service representative within the warranty period.
- The product is used only under the conditions that are described in this manual.
- The monitoring equipment that is incorporated in the product is correctly connected and in use.
- All service and repair work that is done by Xylem authorized personnel.
- Genuine Xylem parts are used.

Limitations

The warranty does not cover defects that are caused by these situations:

- Deficient maintenance
- Improper installation
- Modifications or changes to the product and installation that are made without consulting a Xylem authorized representative
- Incorrectly executed repair work
- Normal wear and tear

Xylem assumes no liability for these situations:

- Bodily injuries
- Material damages
- Economic losses

Warranty claim

Xylem products are high-quality products with expected reliable operation and long life. However, should the need for a warranty claim arise, contact your local sales and service representative.

2 Transportation and Storage

2.1 Inspect the delivery

1. Check the outside of the package for evident signs of damage.
2. Notify our distributor within eight days of the delivery date, if the product bears visible signs of damage.

Unpack the unit

1. Follow applicable step:
   - If the unit is packed in a carton, then remove the staples and open the carton.
   - If the unit is packed in a wooden crate, then open the cover while paying attention to the nails and straps.
2. Remove the securing screws or the straps from the wooden base.

2.1.1 Examine the unit

1. Remove packing materials from the product.

   Dispose of all packing materials in accordance with local regulations.
2. To determine whether any parts have been damaged or are missing, examine the product.
3. If applicable, unfasten the product by removing any screws, bolts, or straps.

   Use care around nails and straps.
4. If there is any issue, then contact a sales representative.

2.2 Lifting guidelines

Precautions



WARNING:

- Observe accident prevention regulations in force.
- Crush hazard. The unit and the components can be heavy. Use proper lifting methods and wear steel-toed shoes at all times.

Check the gross weight that is indicated on the package in order to select proper lifting equipment.

Position and fastening

The pump or pump unit can be transported only horizontally. Make sure that the pump or pump unit is securely fastened during transportation and cannot roll or fall over.



WARNING:

Do not use eyebolts screwed on the motor for handling the whole electric pump unit.

Do not use the shaft end of the pump or of the motor to handle the pump, the motor or the unit.

- Eyebolts screwed onto the motor may be exclusively used to handle the individual motor or, in case of a not balanced distribution of weights, to partially lift the unit vertically starting from a horizontal displacement.

Pump unit must always be fixed and transported as shown in *Figure 1* on page 9, and the pump without motor must be fixed and transported as shown in *Figure 2* on page 9, *Figure 3* on page 9, and *Figure 4* on page 10.

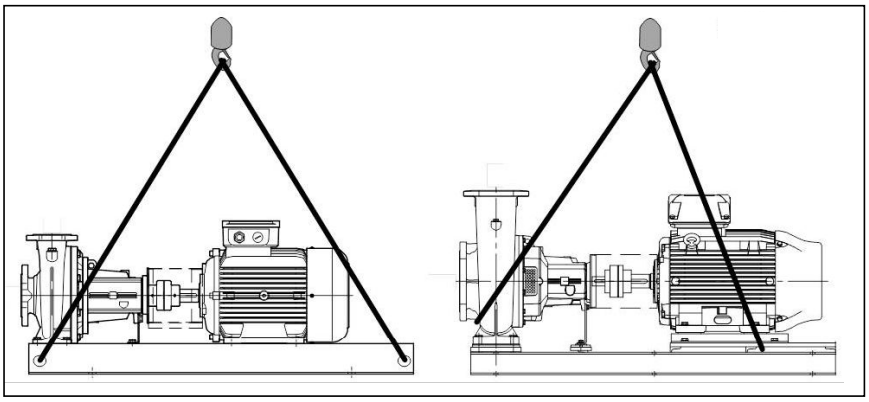


Figure 1

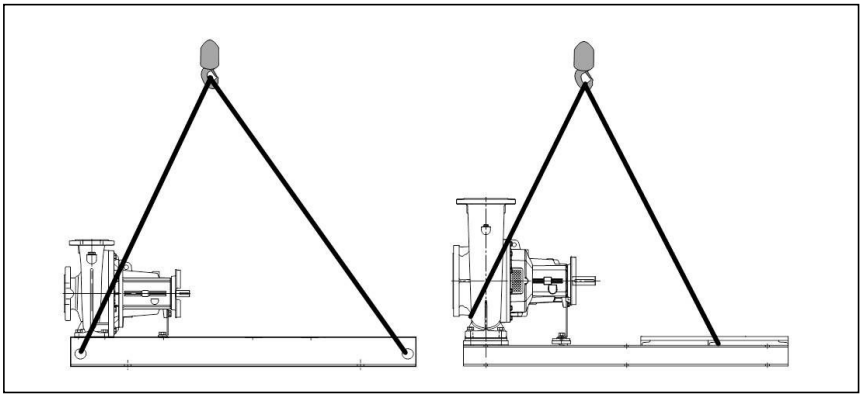


Figure 2

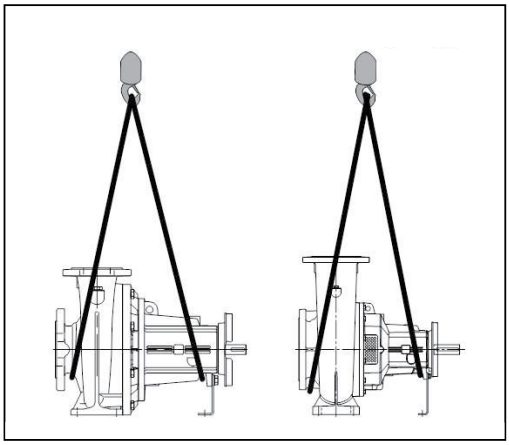


Figure 3

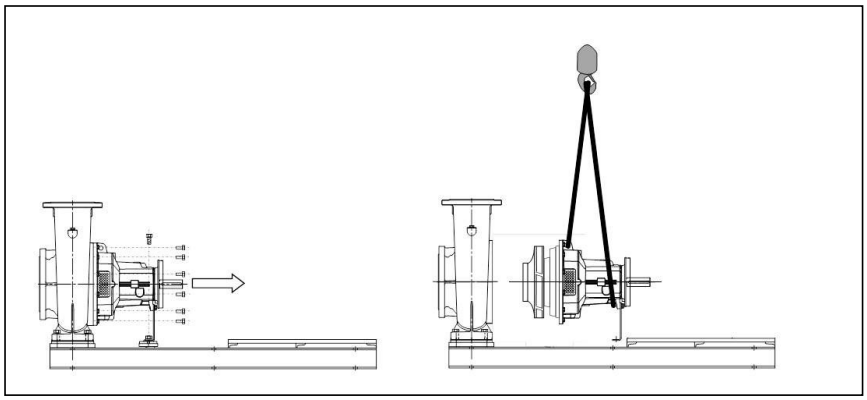


Figure 4

2.3 Storage guidelines

Storage location

The product must be stored in a covered and dry location free from heat, dirt, and vibrations.

NOTICE:

Protect the product against humidity, heat sources, and mechanical damage.

NOTICE:

Do not place heavy weights on the packed product.

2.3.1 Long-term storage

If the unit is stored for more than 6 months, these requirements apply:

- Store in a covered and dry location.
- Store the unit free from heat, dirt, and vibrations.
- Rotate the pump shaft by hand several times at least every three months.

Refer to the drive unit and coupling manufacturers for their long-term storage procedures.

For questions about possible long-term storage treatment services, please contact your local sales and service representative.

Ambient temperature

The product must be stored at an ambient temperature from -5°C to +40°C (23°F to 104°F).

3 Product Description

3.1 General description

1710 Series is foot mounted, single stage and end suction pump with back pull-out design and is in accordance with the international standard ISO 2858.

It comes in frame mounted series applicable for flexible coupling (optional Flexible spacer coupling) and Stub shaft coupling. Bare shaft option to be coupled to any standard IEC motor to substitute an existing ISO 2858 pump.

3.2 Pump application

Designed for non-combustible, low viscosity liquids, free of solids, abrasive materials, and fibers. Different chemical compositions are capable of being pumped by using different construction and seal options.

This highly efficient and flexible design makes the 1710 Series pump ideal for a wide range of pumping applications, including non-listed fire pump, HVAC, Industrial secondary process, Water transfer and more.

Operating conditions:

- Flowrate: max 2100 m³/h
- Pump head: max 160 m
- Fluid temp: -15°C to 120°C
- Max rating: 355 kW

For other applications, contact your local sales and service representative.

Improper use



WARNING:

Improper use of the pump may create dangerous conditions and cause personal injury and damage to property.

An improper use of the product leads to the loss of the warranty.

Examples of improper use:

- Liquids not compatible with the pump construction materials
- Hazardous liquids (such as toxic, explosive, flammable, or corrosive liquids)
- Potable liquids other than water (for example, wine or milk)

Examples of improper installation:

- Hazardous locations (such as explosive, or corrosive atmospheres).
- Location where the air temperature is very high or there is poor ventilation.
- Outdoor installations where there is no protection against rain or freezing temperatures.



DANGER:

Do not use this pump to handle flammable and/or explosive liquids.

NOTICE:

- Do not use the pump for flow rates beyond the specified flow rates on the data plate.

Special applications

Contact the local sales and service representative in the following cases:

- If the density and/or viscosity value of the pumped liquid exceeds the value of water, such as water with glycol; as it may require a more powerful motor.
- If the pumped liquid is chemically treated (for example softened, deionized, demineralized etc.).
- Any situation that is different from the ones that is described and relate to the nature of the liquid.

3.3 Pump denomination and construction features

Identification Code

1	2	3 4 5	6	7	8	9	10	11	12	13	14	15	16	17	18
1710	F	125 - 100 - 200	G	45kW	2P	/A	3	F	A	A	A	C	T	4	N

1. Pump Type (4 digits)
   1710= Series name
2. Coupling (1 digit)
   [F] = Flexible coupling frame mounted
   [C] = Flexible spacer coupling frame mounted
   [S] = Stub Shaft Coupling
   [ ] = Bare Shaft
3. Suction Size in mm (2~3 digits)
4. Discharge Size in mm (2~3 digits)
5. Nominal Impeller Size in mm (3 digits)
6. Lubrication (1 digit)
   [G]* = Greased Lubricated
   [O]* = Oil Lubricated
   [ ]** = NA
7. Motor rating kW (3~6 digits)
8. Number of Motor Poles (2 digits)
   2P
   4P
   6P*
9. Motor brand (1 digit)
   [A] = Siemens
   [B] = Other
   [C]*** = NA
10. Motor efficiency class (1 digit)
    [3] = IE3
    [4] = IE4
    [0]*** = NA
11. Motor Insulation class (1 digit)
    [F] = F Class
    [H] = H Class
    [X]*** = NA
12. Motor winding protection code (1 digit)
    [A] = No
    [B] = PTC
    [C] = PT100
    [X]*** = NA
13. Motor Anticondensation heater (1 digit)
    [A] = No
    [B] = Yes
    [X]*** = NA
14. Motor Separately Driver Fan (1 digit)
    [A] = No
    [B] = Yes
    [X]*** = NA
15. Casing Material (1 digit)
    [C] = Cast Iron
    [D] = Ductile Iron
    [N] = SS316
    [T] = SS304
    [R] = Super Duplex
16. Impeller Material (1 digit)
    [B] = Bronze
    [N] = SS316
    [T] = SS304
    [R] = Super Duplex
17. Seal material (1 digit)
    [4] = Carbon/Ceramic/FPM
    [Z] = SiC/SiC/FPM
    [P]* = Gland Packing
18. Working Pressure (1 digit)
    [N] = PN16
    [H] = PN25

* Available for Coupling option [F] & [C]

** Available for Coupling option [S]

***Available for Coupling option [ ]

COMPONENT	STANDARD OPTION	OTHER AVAILABLE OPTION			
Casing	Cast Iron	Ductile Iron	304 SS	316 SS	Super Duplex
Shaft (seal on shaft)	420 SS	Duplex SS			
Shaft (seal on sleeve)*	420 SS	4135			
Impeller	304 SS	Bronze C87600	316 SS	Super Duplex	
Casing Wear Rings	Bronze C90300	304 SS	316 SS	Super Duplex	
Mechanical Seal	Carbon/ Ceramic/FPM	SiC/SiC/FPM			
Shaft Sleeves*	304 SS	316 SS		Duplex SS	
Shaft sleeve nuts*	304 SS	316 SS		Duplex SS	
Packing*	PTFE	—			
Lantern ring*	304 SS	—			
Seal flush lines*	304 SS	316 SS			

*Note: Available for Gland Packing (gland packing is an optional against mechanical seal).

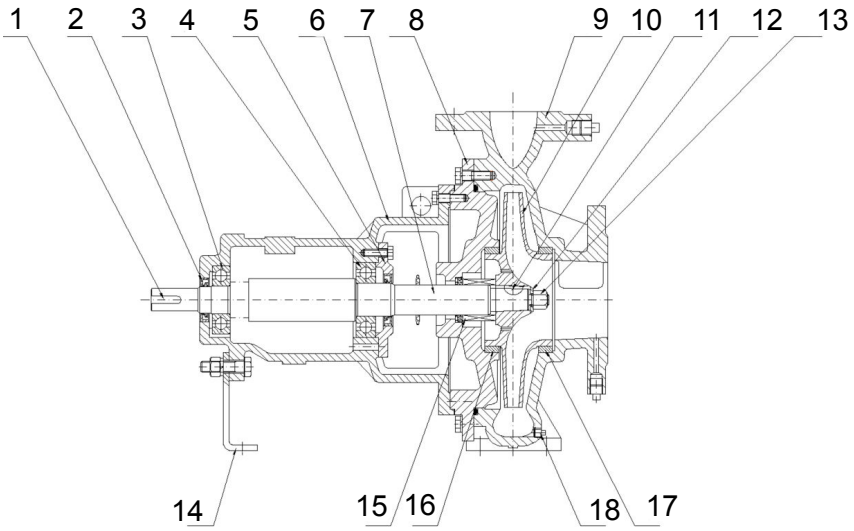
Large choice of casing and impeller material combinations, both PN16 and PN25, wide range of shaft sealing solutions to solve different needs in the industrial market.

Flanges

The flanges are drilled to EN1092/ISO7005, PN16. To enhance versatility, we can deliver PN25 drilled flanges and ASME B16.5 compatible drilled flanges, Class 150 and 300.

Bearing Housing and Bearings

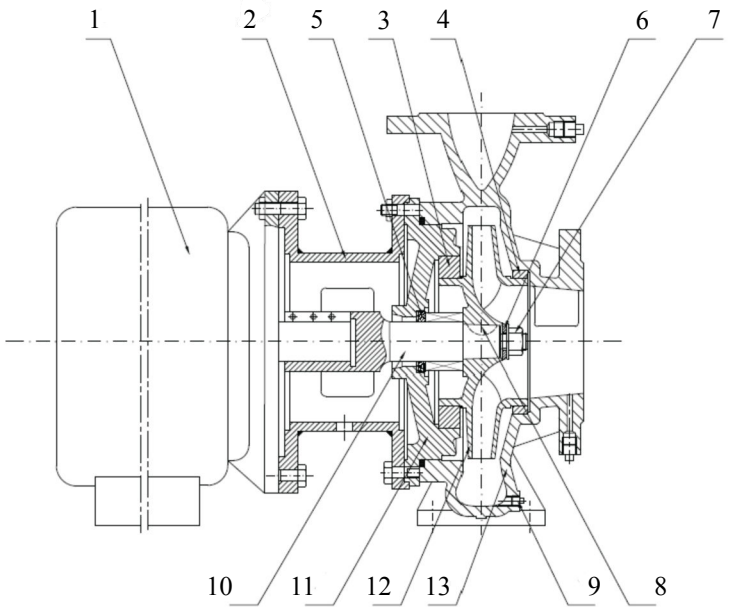
1710 Standard features include a removable bearing cap on the non-drive end and lip seals fitted to both drive and non-drive ends in order to minimize the ingress of foreign liquids and materials. Bearings are normal bearings with pre-packed grease. Alternative bearing manufacturers and lubrication methods are available upon request.



1. Coupling Key	10. Impeller
2. Oil Seal-Drive End	11. Half Round Key
3. Bearing-Driver End	12. Impeller Washer
4. Bearing-Impeller End	13. Impeller Nut
5. Bearing Cover	14. Foot Support
6. Bearing Housing	15. Mechanical Seal
7. Shaft	16. Rear Wear Ring*
8. Back Plate	17. Front Wear Ring
9. Pump Casing	18. Drain Plug

*Note: Group 4, 1710 has no rear wear ring.

Stub shaft construction



1. Motor	8. Impeller Key
2. Motor Adaptor	9. Drain Plug
3. Rear Wear Ring	10. Stub Shaft
4. Front Wear Ring	11. Back Plate
5. Mechanical Seal	12. Impeller
6. Impeller Washer	13. Pump Casing
7. Impeller Nut	

4 Installation

Preinstallation

Before the installation of the pump unit, check whether the packing list matches with the order list and the data that are printed on the nameplate.

Precautions



WARNING:

- Observe accident prevention regulations in force.
- Use suitable equipment and protection.
- Always refer to the local and/or national regulations, legislation, and codes in force regarding the selection of the installation site, plumbing, and power connections.



Electrical Hazard:

- Make sure that all connections are performed by qualified installation technicians and in compliance with the regulations in force.
- Before starting work on the unit, make sure that the unit and the control panel are isolated from the power supply and cannot be energized. This applies to the control circuit as well.

Grounding (earthing)



Electrical Hazard:

- Always connect the external protection conductor to ground (earth) terminal before making other electrical connections.
- You must ground (earth) all electrical equipment. This applies to the pump equipment, the driver, and any monitoring equipment. Test the ground (earth) lead to verify that it is connected correctly.
- If the motor cable is jerked loose by mistake, the ground (earth) conductor should be the last conductor to come loose from its terminal. Make sure that the ground (earth) conductor is longer than the phase conductors. This applies to both ends of the motor cable.
- Add additional protection against lethal shock. Install a high-sensitivity differential switch (30 mA) [residual current device RCD].

4.1 Facility requirements

4.1.1 Pump location



DANGER:

Do not use this unit in environments that may contain flammable/explosive or chemically aggressive gases or powders.

Guidelines

Observe the following guidelines regarding the location of the product:

- Make sure that no obstructions hinder the normal flow of the cooling air that is delivered by the motor fan.
- Make sure that the installation area is protected from any fluid leaks, or flooding.
- If possible, place the pump slightly higher than the floor level.
- The relative humidity of the ambient air must be less than 50% at +40°C (+104°F).
- Contact the Sales and Service Department if:

– The relative air humidity conditions exceed the guidelines.
– The unit is located more than 1000 m (3000 ft) above the sea level. The motor performance may need to be de-rated or replaced with a more powerful motor.

Pump positions and clearance

Provide adequate light and clearance around the pump. Make sure that it is easily accessible for installation and maintenance operations.

4.2 Foundation requirements

Requirements

- The foundation must be able to absorb any type of vibration and form a permanent, rigid support for the unit.
- Provide a flat, substantial concrete foundation in order to prevent strain and distortion when you tighten the foundation bolts.

4.3 Piping requirements

Precautions



WARNING:

- Use pipes suited to the maximum working pressure of the pump. Failure to do so can cause the system to rupture, with the risk of injury.
- Make sure that all connections are performed by qualified installation technicians and in compliance with the regulations in force.

NOTICE:

Observe all regulations issued by authorities having jurisdiction and by companies managing the public water supplies if the pump is connected to a public water system. If required, install appropriate backflow-prevention device on the suction side.

Piping checklist

Check that the following requirements are met:

- All piping is independently supported, piping must not place a burden on the unit.
- Flexible pipes or unions are used, in order to avoid transmission of pump vibrations to the pipes and vice versa.
- Use wide bends, avoid using elbows which cause excessive flow resistance.
- The suction piping is perfectly sealed and airtight.
- If the pump is used in an open circuit, then the diameter of the suction pipe is suited to the installation conditions. The suction pipe must not be smaller than the diameter of the suction port.
- If the suction piping must be larger than the suction side of the pump, then an eccentric pipe reducer is installed.
- If the pump is placed above liquid level, a foot valve is installed at the end of the suction piping.
- The foot valve is fully immersed into the liquid so that air cannot enter through the suction vortex, when the liquid is at the minimum level and the pump is installed above the liquid source.
- Appropriately sized on-off valves are installed on the suction piping and on the delivery piping (downstream to the check valve) for regulation of the pump capacity, for pump inspection, and for maintenance.
- Appropriately sized on-off valves are installed on the delivery piping (downstream to the check valve) for regulation of the pump capacity, for pump inspection, and for maintenance.
- In order to prevent back flow into the pump when pump is turned off a check valve is installed on the delivery piping.



WARNING:

Do not use the on-off valve on the discharge side in the closed position in order to throttle the pump for more than a few seconds. If the pump must operate with the discharge side closed for more than a few seconds, a bypass circuit must be installed to prevent overheating of the liquid inside the pump.

4.4 Electrical requirements

- The local regulations in force overrule these specified requirements.
- In the case of fire fighting systems (hydrants and/or sprinklers), check the local regulations in force

Electrical connection checklist

Check that the following requirements are met:

- The electrical leads are protected from high temperature, vibrations, and collisions.
- The power supply line is provided with:
  - A short-circuit protection device
  - A mains isolator switch with a contact gap of at least 3 mm

The electrical control panel checklist

NOTICE:

The control panel must match the ratings of the electric pump. Improper combinations could fail to guarantee the protection of the motor.

Check that the following requirements are met:

- The control panel must protect the motor against overload and short-circuit.
- Install the correct overload protection (thermal relay or motor protector).
- The control panel must be equipped with a dry-running protection system to which a pressure switch, float switch, probes, or other suitable device is connected.
- The following devices are recommended for use on the suction side of the pump:
  - When the liquid is pumped from a water system, use a pressure switch.
  - When the liquid is pumped from a storage tank or reservoir, use a float switch or probes.
- When thermal relays are used, relays that are sensitive to phase failure are recommended.

4.5 Install the pump

Install the pump set

1. Position the pump set on the foundation and level it with the help of a spirit level that is placed on the discharge port.

   The permissible deviation is 0.2 mm/m.
2. Remove the plugs covering the ports.
3. Align the pump and piping flanges on both sides of the pump. Check the alignment of the bolts.
4. Fasten the piping with bolts to the pump. Do not force the piping into place.
5. Use shims for height compensation, if necessary.

   Always fit shims, if any, immediately to the left and right of the foundation bolts between the baseplate/foundation. For a bolt-to-bolt distance (L) > 800 mm, fit extra shims halfway between the bolt holes.
6. Make sure that all shims lie perfectly flush.
7. Insert the foundation bolts into the holes provided.

8. Use concrete to set the foundation bolts into the foundation.
9. Wait until the concrete has set firmly, and then level the baseplate.
10. Tighten the foundation bolts evenly and firmly.

Note:

- For baseplates, it is recommended to grout the baseplate with low-shrinkage concrete.
- If the transmission of vibrations can be disturbing, provide vibration-damping supports between the pump and the foundation.

Mount the pump to a base frame

Be sure to check that the following are adhered to:

- Solid base frame which does not twist or vibrate during operation (resonance).
- Mounting surfaces of the pump feet and the motor on the base frame must be flat (machining is recommended).
- Safe fastening of pump and motor must be guaranteed.
- Adequate space between pump and motor shaft must be left depending on the used coupling.
- Between pump and base frame must be an adequate shimming, so that in case of replacement the same height between bottom and centerline can be adjusted (recommended vertical adjustment 4-6 mm).

4.6 Piping checklist

Check that the following are adhered to:

- The suction lift line has been laid with a rising slope, at positive suction head line with a downward slope towards the pump.
- The nominal diameters of the pipelines are at least equal to the nominal diameters of the pump ports.
- The pipelines have been anchored in close proximity to the pump and connected without transmitting any stresses or strains.



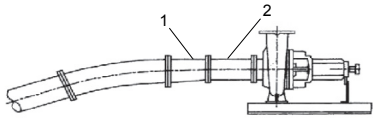
CAUTION:

Welding beads, scale and other impurities in the piping damage the pump.

- Free the piping from any impurities.
- If necessary, install a filter.

4.7 Suction pipeline

Suction pipeline must have no air leaks. Suction piping should be the same size or larger than that of the pump suction flange. Tapered eccentric reducer should be used when by using larger diameter pipe work. see figure

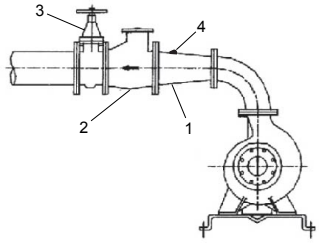


1. Eccentric taper pipe
2. Straight pipe

The pump should be assembled by using a straight pipe of length not less than three times the pipe diameter. Identify that the end of the suction line is sufficiently below the lowest level of the liquid to prevent the formation of vortex, and the consequent entry of air into the suction pipe. Where there is suction lift, it is essential that a reputable foot valve by using a water opening of at least to that of the pipe, is used.

4.8 Discharge pipeline

It is recommended that a non-return valve is assembled to the pumps discharge. The installation of a gate valve after the non-return valve is beneficial to aid in priming of the pump, trimming of the discharge flow and maintenance on the non return valve.



1. Concentric taper pipe
2. Check valve
3. Gate valve
4. Priming plug

Discharge pipeline should be selected of a size applicable to carry the required capacity, such that the friction head created is not too much. When the pipeline is laid over undulating ground by using high points where air pockets can form, vent cocks must be put to expel the accumulation of air, which will affect the discharge capacity of the pump.

4.9 Coupling arrangement

After mounting to the foundation and the connection of the piping, the coupling must be adjusted again, even if the unit was delivered completely mounted on the frame.

The following is intended as a guide only for the correct selection of the appropriate coupling .

- Type of pump
- Type of drive
- The maximum output power/shaft power of the drive
- Duty type: Continuous or Intermittent
- Shaft diameter & key size of pump & drive
- Alignment: Refer to coupling manufactures specifications.

Contact the local sales and service representative for any requests or information.

NOTICE:

Improper alignment of the unit can lead to damages at coupling and unit.

4.10 Electrical installation

1. Remove the screws of the terminal box cover.
2. Connect and fasten the power cables according to the applicable wiring diagram.
   a) Connect the ground (earth) lead.
      Make sure that the ground (earth) lead is longer than the phase leads.
   b) Connect the phase leads.
3. Mount the terminal box cover.

   NOTICE:

   Tighten the cable glands carefully to ensure protection against cable slipping and humidity entering the terminal box.
4. If the motor is not equipped with automatic reset thermal protection, then adjust the overload protection according to the list below.

- If the motor is used with full load, then set the value to the nominal current value of electric pump (data plate)
- If the motor is used with partial load, then set the value to the operating current (for example measured with a current pincer).
- If the pump has a star-delta starting system, then adjust the thermal relay to 58% of the nominal current or operating current (only for three-phase motors).

5 Commissioning, Startup, Operation, and Shutdown

Precautions



WARNING:

- Make sure that the drained liquid does not cause damage or injuries.
- The motor protectors can cause the motor to restart unexpectedly. This could result in serious injury.
- Never operate the pump without the coupling guard correctly installed.



CAUTION:

- The outer surfaces of the pump and motor can exceed 40ºC (104ºF) during operation. Do not touch with any part of the body without protective gear.
- Do not put any combustible material near the pump.

NOTICE:

- Never operate the pump below the minimum rated flow, when dry, or without prime.
- Never operate the pump with the delivery ON-OFF valve closed for longer than a few seconds.
- Never operate the pump with the suction ON-OFF valve closed.
- Do not expose an idle pump to freezing conditions. Drain all liquid that is inside the pump. Failure to do so can cause liquid to freeze and damage the pump.
- The sum of the pressure on the suction side (mains, gravity tank) and the maximum pressure that is delivered by the pump must not exceed the maximum working pressure that is allowed (nominal pressure PN) for the pump.
- Do not use the pump if cavitation occurs. Cavitation can damage the internal components.

5.1 Fill the pump

Installations with liquid level above the pump (suction head)

1. Close the on-off valve located downstream from the pump.
2. Remove the fill or gauge plug and open the on/off valve upstream until the water flows out of the hole.
3. Close the fill or gauge plug

Installations with liquid level below the pump (suction lift)

1. All pipe system empty:
   a) Open the on-off valve located upstream from the pump.
   b) Remove the fill plug and the gauge plug. Use a funnel to fill the pump through the fill hole until the water flows out of this hole.
   c) Tighten the fill plug and the gauge plug.
2. Filled discharge pipe system:
   a) Open the on-off valve located upstream from the pump and open the on-off valve downstream.
   b) Remove the gauge plug until water flows out of this hole.
   c) Tighten the gauge plug.

5.2 Check the rotation direction (three-phase motor)

Follow this procedure before start-up.

1. Locate the arrows on the adaptor or the motor fan cover to determine the correct rotation direction.
2. Start the motor.
3. Quickly check the direction of rotation through the coupling guard or through the motor fan cover.
4. Stop the motor.
5. If the rotation direction is incorrect, then do as follows:
   a) Disconnect the power supply.
   b) In the terminal board of the motor or in the electric control panel, exchange the position of two of the three wires of the supply cable.
   c) Check the direction of rotation again.

5.3 Pump operation precautions

General considerations



CAUTION:

- Vary the capacity with the regulating valve in the discharge line. Never throttle the flow from the suction side since this can result in decreased performance, unexpected heat generation, and equipment damage.
- Do not overload the driver. Driver overload can result in unexpected heat generation and equipment damage. The driver can overload in these circumstances:
  - The specific gravity of the pumped fluid is greater than expected.
  - The pumped fluid exceeds the rated flow rate.
- Make sure to operate the pump at or near the rated conditions. Failure to do so can result in pump damage from cavitation or recirculation.

Operation at reduced capacity



WARNING:

Never operate any pumping system with a blocked suction and discharge. Operation, even for a brief period under these conditions, can cause confined pumped fluid to overheat, which results in a violent explosion. You must take all necessary measures to avoid this condition.

NOTICE:

- Avoid increased radial load. Failure to do so can cause stress on the shaft and bearings.
- Avoid heat build-up. Failure to do so can cause rotating parts to score or seize.
- Avoid cavitation. Failure to do so can cause damage to the internal surfaces of the pump.

Operation under freezing conditions

NOTICE:

Do not expose an idle pump to freezing conditions. Drain all liquid that is inside the pump. Failure to do so can cause liquid to freeze and damage the pump.

5.4 Start the pump

The responsibility for checking the correct flow and the temperature of the pumped liquid rests with the installer or owner.

Before starting the pump, make sure that:

- The pump is correctly connected to the power supply.
- The pump is correctly filled according to instructions in *Fill the pump* (chapter 5).
- The on-off valve located downstream from the pump is closed.

1. Start the motor.
2. Gradually open the on-off valve on the discharge side of the pump.
   At the expected operating conditions, the pump must run smoothly and quietly. If not, refer to *Troubleshooting* on page 26.

6 Maintenance

Precautions



Electrical Hazard:

Disconnect and lock out electrical power before installing or servicing the unit.



WARNING:

- Maintenance and service must be performed by skilled and qualified personnel only.
- Observe accident prevention regulations in force.
- Use suitable equipment and protection.
- Make sure that the drained liquid does not cause damage or injuries.

6.1 Service

If the user wishes to schedule regular maintenance deadlines, they are dependent on the type of pumped liquid and on the operating conditions of the pump.

Contact the local sales and service representative for any requests or information regarding routine maintenance or service.

Extraordinary maintenance may be necessary to clean the liquid end and/or replace worn parts.

Pumps with greased for life bearings

Pumps with greased for life bearings do not require any scheduled routine maintenance.

Pumps with re-greaseable bearings

- Regrease at 4000 operating hours, but at least once per year. Clean lubrication nipples (SN) first.
- Use NLGI Grade 2 grease or equivalent.

Contact the local sales and service representative for any requests or information.

Motor bearings

After approximately five years, the grease in the motor bearings is so aged that a replacement of the bearings is recommended. The bearings must be replaced after 25000 operating hours or according to the motor supplier maintenance instructions, whichever is shorter.

Motor with regreasable bearings

Follow motor supplier maintenance instructions.

Coupling

Check the clearance in the coupling elements regularly, at least once a year. We recommend checking every 1000 operating hours or every three months, whichever comes first.

6.2 Inspection checklist

Check the coupling	Check the flexible elements of the coupling. Replace the relevant parts if there is any sign of wear and check the alignment.
Check the mechanical seal	Check for leakage of the mechanical seal. Replace the mechanical seal if leakage is found.

Checking the bearing seals	Check correct seating of axial seal rings mounted on the shaft. Only gentle contact of the sealing lip shall be established.
Check for quiet running	Check frequently for quiet running of the pump with vibration measurements tools.

Check for leakage of the mechanical seal. Replace the mechanical seal if leakage is found.

6.3 Disassemble and replace the pump parts

For more information about spare parts and assembly and disassembly of the pump, contact the local sales and service representative.

7 Troubleshooting

7.1 Troubleshooting for users

The main switch is on, but the electric pump does not start.

Cause	Remedy
The thermal protector incorporated in the pump (if any) has tripped.	Wait until the pump has cooled down. The thermal protector will automatically reset.
The protective device against dry running has tripped.	Check the liquid level in the tank, or the mains pressure.

The electric pump starts, but the thermal protection trips a varying time after.

Cause	Remedy
There are foreign objects (solids or fibrous substances) inside the pump which have jammed the impeller.	Contact the Sales and Service Department.
The pump is overloaded because it is pumping liquid that is too dense and viscous.	Check the actual power requirements based on the characteristics of the pumped liquid and then contact the Sales and Service Department.

The pump runs but delivers too little or no liquid.

Cause	Remedy
The pump is clogged.	Contact the Sales and Service Department.

The troubleshooting instructions in the tables below are for installers only.

7.2 The main switch is on, but the electric pump does not start

Cause	Remedy
There is no power supply.	• Restore the power supply. • Make sure all electrical connections to the power supply are intact.
The thermal protector incorporated in the pump (if any) has tripped.	Wait until the pump has cooled down. The thermal protector will automatically reset.
The thermal relay or motor protector in the electric control panel has tripped.	Reset the thermal protection.
The protective device against dry running has tripped.	Check the: • liquid level in the tank, or the mains pressure. • protective device and its connecting cables.
The fuses for the pump or auxiliary circuits are blown.	Replace the fuses.

7.3 The electric pump starts, but the thermal protector trips or the fuses blow immediately after

Cause	Remedy
The power supply cable is damaged.	Check the cable and replace as necessary.
The thermal protection or fuses are not suited for the motor current.	Check the components and replace as necessary.

Cause	Remedy
The electric motor is short circuit.	Check the components and replace as necessary.
The motor overloads.	Check the operating conditions of the pump and reset the protection.

7.4 The electric pump starts, but the thermal protector trips or the fuses blow a short time after

Cause	Remedy
The electrical panel is situated in an excessively heated area or is exposed to direct sunlight.	Protect the electrical panel from heat source and direct sunlight.
The power supply voltage is not within the working limits of the motor.	Check the operating conditions of the motor.
A power phase is missing.	Check the • power supply • electrical connection

7.5 The electric pump starts, but the thermal protector trips a varying time after

Cause	Remedy
There are foreign objects (solids or fibrous substances) inside the pump which have jammed the impeller.	Contact the local sales and service representative.
The pumps delivery rate is higher than the limits specified on the data plate.	Partially close the ON-OFF valve down stream until the delivery rate is equal or less than the limits specified on the data plate.
The pump is overloaded because it is pumping liquid that is too dense and viscous.	Check the actual power requirements based on the characteristics of the pumped liquid and replace the motor accordingly.
The motor bearings are worn.	Contact the local sales and service representative.

7.6 The electric pump starts, but the system's general protection is activated

Cause	Remedy
A short circuit in the electrical system.	Check the electrical system.

7.7 The electric pump starts, but the system's residual current device (RCD) is activated

Cause	Remedy
There is an ground (earth) leakage.	Check the insulation of the electrical system components.

7.8 The pump runs but delivers too little or no liquid

Cause	Remedy
There is air inside the pump or the piping.	• Bleed the air

Cause	Remedy
The pump is not correctly primed.	Stop the pump and repeat the prime procedure. If the problem continues: • Check that the mechanical seal is not leaking. • Check the suction pipe for perfect tightness. • Replace any valves that are leaking.
The throttling on the delivery side is too extensive.	Open the valve.
Valves are locked in closed or partially closed position.	Disassemble and clean the valves.
The pump is clogged.	Contact the local sales and service representative.
The piping is clogged.	Check and clean the pipes.
The rotation direction of the impeller is wrong. (three-phase version)	Change the position of two of the phases on the terminal board of the motor or in the electric control panel.
The suction lift is too high or the flow resistance in the suction pipes is too great.	Check the operating conditions of the pump. If necessary, do the following: • Decrease the suction lift • Increase the diameter of the suction pipe.

7.9 The electric pump stops, and then rotates in the wrong direction

Cause	Remedy
There is a leakage in one or both of the following components: • The suction pipe • The foot valve or the check valve	Repair or replace the faulty component.
There is air in the suction pipe.	Bleed the air.

7.10 The pump starts up too frequently

Cause	Remedy
There is a leakage in one or both of the following components: • The suction pipe • The foot valve or the check valve	Repair or replace the faulty component.
There is a ruptured membrane or no air pre-charge in the pressure tank.	See the relevant instructions in the pressure tank manual.

7.11 The pump vibrates and generates too much noise

Cause	Remedy
Pump cavitation	Reduce the required flow rate by partially closing the ON-OFF valve downstream from the pump. If the problem persists check the operating conditions of the pump (for example height difference, flow resistance, liquid temperature).
The motor bearings are worn.	Contact the local sales and service representative.
There are foreign objects inside the pump.	Contact the local sales and service representative.
Impeller rubs on the wear ring	Contact the local sales and service representative.
Coupling misaligned	Check the coupling alignment.
Flexible elements of the coupling worn	Check and replace the relevant parts if there is any sign of wear.

For any other situation, refer to the local sales and service representative.

Xylem |’zīləm|

1) The tissue in plants that brings water upward from the roots;
2) a leading global water technology company.

We’re a global team unified in a common purpose: creating advanced technology solutions to the world’s water challenges. Developing new technologies that will improve the way water is used, conserved, and re-used in the future is central to our work. Our products and services move, treat, analyze, monitor and return water to the environment, in public utility, industrial, residential and commercial building services settings. Xylem also provides a leading portfolio of smart metering, network technologies and advanced analytics solutions for water, electric and gas utilities. In more than 150 countries, we have strong, long-standing relationships with customers who know us for our powerful combination of leading product brands and applications expertise with a strong focus on developing comprehensive, sustainable solutions.

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